A covariant energy density functional derived from chiral nuclear forces at NLO

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Outline

- Introduction
- Theoretical Framework
  - Relativistic Chiral NN Force
  - Nuclear Matter
  - Spherical Even-even Nuclei
- Numerical Details
- Results and Discussion
- Summary

Introduction

- Nuclear DFT is uniquely global tool:
  - Nuclear Ground States, Excited States and Nuclear Dynamics.

    Meng, J., Zhao, P. Relativistic density functional theory in nuclear physics. AAPPS Bull. 31, 2 (2021)
  - Low-lying excited states of odd-mass nuclei, and applications related to nuclear Schiff moment calculations.

    E. F. Zhou, X. Y. Wu, and J. M. Yao Phys. Rev. C 109, 034305(2024)
  - $0\nu\beta\beta$ matrix elements.

    J. M. Yao, L. S. Song et al. Phys. Rev. C 91, 024316 (2015)
- Drawbacks:
  - Different EDFs leading to strong model dependence and theoretical uncertainties.
  - Lack the capability to analyze truncation errors.
- Our work:
  - Construct a covariant energy density functional derived from chiral nuclear forces.

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Relativistic Chiral NN Force

The form of Relativistic NN contact Lagrangian[1]:

$$\mathcal{L} = \sum_i c_i O_i \left(\{\psi\}\right)$$

where operator O_i takes the form:

$$\frac{1}{(2m_n)^{N_d}} \left(\bar{\psi} i \overleftrightarrow{\partial}^{\alpha} i \overleftrightarrow{\partial}^{\beta} \cdots \Gamma_A \psi\right) \partial^{\lambda} \partial^{\mu} \cdots \left(\bar{\psi} i \overleftrightarrow{\partial}^{\sigma} i \overleftrightarrow{\partial}^{\tau} \cdots \Gamma_B \psi\right)$$

and $\Gamma \in \{\mathbf{1}, \gamma_5, \gamma^\mu, \gamma_5\gamma^\mu, \sigma^{\mu\nu}, g^{\mu\nu}, \epsilon^{\mu\nu\rho\sigma}\}$.

Define the chiral order (Relativistic Power Counting) of each building block:

Building blocks	$\mathbf{1}$	γ_5	γ_μ	$\gamma_5\gamma_\mu$	$\sigma_{\mu\nu}$	$\epsilon_{\mu\nu\rho\sigma}$	$\overleftrightarrow{\partial}_\mu$	∂_μ
$\mathcal{P}$	+	-	+	-	+	-	+	+
$\mathcal{C}$	+	+	-	+	-	+	-	+
h.c.	+	-	+	+	+	+	-	+
Chiral order	0	1	0	0	0	-	0	1

Relativistic NN contact Lagrangian up to NLO:

$\mathcal{O}(q^0)$	$\tilde{O}_1$	$\left(\bar{\psi}\psi\right)\left(\bar{\psi}\psi\right)$	$\mathcal{O}(q^2)$	$\tilde{O}_{10}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\psi\right)\partial^2\left(\bar{\psi}\psi\right)$
	$\tilde{O}_2$	$\left(\bar{\psi}\gamma^\mu\psi\right)\left(\bar{\psi}\gamma_\mu\psi\right)$		$\tilde{O}_{11}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\gamma_\mu\psi\right)$
	$\tilde{O}_3$	$\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)$		$\tilde{O}_{12}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\gamma_5\gamma_\mu\psi\right)$
	$\tilde{O}_4$	$\left(\bar{\psi}\sigma^{\mu\nu}\psi\right)\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$		$\tilde{O}_{13}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}\psi\right)\partial^2\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$
$\mathcal{O}(q^2)$	$\tilde{O}_5$	$\left(\bar{\psi}\gamma_5\psi\right)\left(\bar{\psi}\gamma_5\psi\right)$		$\tilde{O}_{14}$	$\frac{1}{4m_n^2}\left(\bar{\psi} i \overleftrightarrow{\partial}^{\alpha}\psi\right)\left(\bar{\psi} i \overleftrightarrow{\partial}_{\alpha}\psi\right) - \tilde{O}_1$
	$\tilde{O}_6$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i \overleftrightarrow{\partial}^{\alpha}\psi\right)\left(\bar{\psi}\gamma_5\gamma_\alpha i \overleftrightarrow{\partial}_{\mu}\psi\right)$		$\tilde{O}_{15}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu i \overleftrightarrow{\partial}^{\alpha}\psi\right)\left(\bar{\psi}\gamma_\mu i \overleftrightarrow{\partial}_{\alpha}\psi\right) - \tilde{O}_2$
	$\tilde{O}_7$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu} i \overleftrightarrow{\partial}^{\alpha}\psi\right)\left(\bar{\psi}\sigma_{\mu\alpha} i \overleftrightarrow{\partial}_{\nu}\psi\right)$		$\tilde{O}_{16}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i \overleftrightarrow{\partial}^{\alpha}\psi\right)\left(\bar{\psi}\gamma_5\gamma_\mu i \overleftrightarrow{\partial}_{\alpha}\psi\right) - \tilde{O}_3$
	$\tilde{O}_8$	$\frac{1}{4m_n^2}\left(\bar{\psi} i \overleftrightarrow{\partial}^{\mu}\psi\right)\partial^\nu\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$		$\tilde{O}_{17}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu} i \overleftrightarrow{\partial}^{\alpha}\psi\right)\left(\bar{\psi}\sigma_{\mu\nu} i \overleftrightarrow{\partial}_{\alpha}\psi\right) - \tilde{O}_4$
	$\tilde{O}_9$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\alpha}\psi\right)\partial_\alpha\partial^\nu\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$			

[1] Yang Xiao, Li-Sheng Geng, and Xiu-Lei Ren, PHYSICAL REVIEW C 99, 024004 (2019)

Theoretical Framework

We start with the Largrangian density:

$$\mathcal{L} = \mathcal{L}^{\text{free}} + \mathcal{L}^{\chi\text{EFT}} + \mathcal{L}^{\text{IV}} + \mathcal{L}^{\text{em}}$$

where

$$\begin{aligned}
\mathcal{L}^{\text{free}} =& \bar{\psi}\left(i\gamma^{\mu}\partial_{\mu} - m\right)\psi \\
\mathcal{L}^{\chi\text{EFT}} =& -\frac{1}{2}\sum_{i=1}^{17} c_i \tilde{O}_i \\
\mathcal{L}^{\text{IV}} =& -\frac{1}{2}c_{1t}\left(\bar{\psi}\vec{\tau}\psi\right)\left(\bar{\psi}\vec{\tau}\psi\right) - \frac{1}{2}c_{2t}\left(\bar{\psi}\vec{\tau}\gamma^{\mu}\psi\right)\left(\bar{\psi}\vec{\tau}\gamma_{\mu}\psi\right) \\
& -\frac{1}{2}\frac{c_{10t}}{4m^2}\left(\bar{\psi}\vec{\tau}\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\psi\right) \\
& -\frac{1}{2}\frac{c_{11t}}{4m^2}\left(\bar{\psi}\vec{\tau}\gamma^{\mu}\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\gamma_{\mu}\psi\right) \\
\mathcal{L}^{\text{em}} =& -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - e\frac{1-\tau_3}{2}\bar{\psi}\gamma^{\mu}\psi A_{\mu}
\end{aligned}$$

Rearrange $\Rightarrow$

$$\mathcal{L} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{em}} + \mathcal{L}_{\text{LO}} + \mathcal{L}_{\text{NLO}}$$

where

$$\begin{aligned}
\mathcal{L}_{\text{LO}} =& -\frac{1}{2}\sum_{i=1}^{4} c_i \tilde{O}_i - \frac{1}{2}c_{1t}\left(\bar{\psi}\vec{\tau}\psi\right)\left(\bar{\psi}\vec{\tau}\psi\right) \\
& -\frac{1}{2}c_{2t}\left(\bar{\psi}\vec{\tau}\gamma^{\mu}\psi\right)\left(\bar{\psi}\vec{\tau}\gamma_{\mu}\psi\right) \\
\mathcal{L}_{\text{NLO}} =& -\frac{1}{2}\sum_{i=5}^{17} c_i \tilde{O}_i - \frac{1}{2}\frac{c_{10t}}{4m^2}\left(\bar{\psi}\vec{\tau}\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\psi\right) \\
& -\frac{1}{2}\frac{c_{11t}}{4m^2}\left(\bar{\psi}\vec{\tau}\gamma^{\mu}\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\gamma_{\mu}\psi\right)
\end{aligned}$$

The complete Lagrangian contains **6+15=21** coupling constants.

Nuclear Matter

The Lagrangian in Nuclear Matter takes the form:

$$\mathcal{L} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{LO}} + \mathcal{L}_{\text{NLO}}$$

where

$$\begin{aligned}\mathcal{L}_{\text{LO}} =& -\frac{1}{2}(c_1 - c_{14})\left(\bar{\psi}\psi\right)\left(\bar{\psi}\psi\right) - \frac{1}{2}(c_2 - c_{15})\left(\bar{\psi}\gamma^\mu\psi\right)\left(\bar{\psi}\gamma_\mu\psi\right) \\ & -\frac{1}{2}c_{1t}\left(\bar{\psi}\vec{\tau}\psi\right)\left(\bar{\psi}\vec{\tau}\psi\right) - \frac{1}{2}c_{2t}\left(\bar{\psi}\vec{\tau}\gamma^\mu\psi\right)\left(\bar{\psi}\vec{\tau}\gamma_\mu\psi\right) \\ \mathcal{L}_{\text{NLO}} =& -\frac{1}{2}\frac{c_{14}}{4m^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\mu\psi\right)\left(\bar{\psi}i\overleftrightarrow{\partial}_\mu\psi\right) \\ & -\frac{1}{2}\frac{c_{15}}{4m^2}\left(\bar{\psi}i\gamma^\nu\overleftrightarrow{\partial}^\mu\psi\right)\left(\bar{\psi}i\gamma_\nu\overleftrightarrow{\partial}_\mu\psi\right)\end{aligned}$$

Define[2]

$$\begin{aligned}\rho_S &= \left\langle\bar{\psi}\psi\right\rangle & \vec{j}^{\,\mu}_{tV} &= \left\langle\bar{\psi}\vec{\tau}\gamma^\mu\psi\right\rangle \\ \vec{\rho}_{tS} &= \left\langle\bar{\psi}\vec{\tau}\psi\right\rangle & j^\mu_{\text{d}} &= \tfrac{1}{2}\left\langle\bar{\psi}i\overleftrightarrow{\partial}^\mu\psi\right\rangle \\ j^\mu_V &= \left\langle\bar{\psi}\gamma^\mu\psi\right\rangle & j^{\nu\mu}_{\text{d}} &= \tfrac{1}{2}\left\langle\bar{\psi}\gamma^\nu i\overleftrightarrow{\partial}^\mu\psi\right\rangle\end{aligned}$$

The Dirac equation takes the following form:

$$\left[i\gamma^\mu\partial_\mu - \Sigma^\mu i\partial_\mu - (m+S) - \gamma^0 V_0\right]\psi = 0$$

where

$$\begin{aligned}S &= (c_1 - c_{14})\,\rho_S + c_{1t}\vec{\rho}_{tS}\tau_3, \\ V_0 &= (c_2 - c_{15})\,\rho_V + c_{2t}\vec{\rho}_{tV}\tau_3, \\ \Sigma^\mu &= \frac{1}{m^2}\left(c_{14}j^\mu_{\text{d}} + c_{15}j^{\nu\mu}_{\text{d}}\gamma_\nu\right)\end{aligned}$$

$\mathcal{O}(q^0)$	$\tilde{O}_1, \tilde{O}_{1t}$	$\left(\bar{\psi}\psi\right)\left(\bar{\psi}\psi\right), \left(\bar{\psi}\vec{\tau}\psi\right)\left(\bar{\psi}\vec{\tau}\psi\right)$	$\mathcal{O}(q^2)$	$\tilde{O}_{10}, \tilde{O}_{10t}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\psi\right)\partial^2\left(\bar{\psi}\psi\right), \frac{1}{4m_n^2}\left(\bar{\psi}\vec{\tau}\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\psi\right)$
	$\tilde{O}_2, \tilde{O}_{2t}$	$\left(\bar{\psi}\gamma^\mu\psi\right)\left(\bar{\psi}\gamma_\mu\psi\right), \left(\bar{\psi}\vec{\tau}\gamma^\mu\psi\right)\left(\bar{\psi}\vec{\tau}\gamma_\mu\psi\right)$		$\tilde{O}_{11}, \tilde{O}_{11t}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\gamma_\mu\psi\right), \frac{1}{4m_n^2}\left(\bar{\psi}\vec{\tau}\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\gamma_\mu\psi\right)$
	$\tilde{O}_3$	$\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)$		$\tilde{O}_{12}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\gamma_5\gamma_\mu\psi\right)$
	$\tilde{O}_4$	$\left(\bar{\psi}\sigma^{\mu\nu}\psi\right)\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$		$\tilde{O}_{13}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}\psi\right)\partial^2\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$
$\mathcal{O}(q^2)$	$\tilde{O}_5$	$\left(\bar{\psi}\gamma_5\psi\right)\left(\bar{\psi}\gamma_5\psi\right)$		$\tilde{O}_{14}$	$\frac{1}{4m_n^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_1$
	$\tilde{O}_6$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_5\gamma_\alpha i\overleftrightarrow{\partial}_\mu\psi\right)$		$\tilde{O}_{15}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_\mu i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_2$
	$\tilde{O}_7$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\sigma_{\mu\alpha}i\overleftrightarrow{\partial}_\nu\psi\right)$		$\tilde{O}_{16}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_5\gamma_\mu i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_3$
	$\tilde{O}_8$	$\frac{1}{4m_n^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\mu\psi\right)\partial^\nu\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$		$\tilde{O}_{17}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\sigma_{\mu\nu}i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_4$
	$\tilde{O}_9$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\alpha}\psi\right)\partial_\alpha\partial^\nu\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$			

[2] S. Typel, T. v. Chossy, and H. H. Wolter, Physical Review C 67(2003)

Nuclear Matter

Consider the momentum eigenstate $\psi(x) = u(p)\, e^{-ip\cdot x}$, Dirac equation can be expressed as a matrix equation:

$$\begin{pmatrix} M^\star - E^\star & \boldsymbol{\sigma}\cdot\mathbf{p}^\star \\ \boldsymbol{\sigma}\cdot\mathbf{p}^\star & -M^\star - E^\star \end{pmatrix} u = 0$$

where

$$E^\star \equiv \left(1 - \frac{c_{15}}{m^2} j_{\mathrm{d}}^{00}\right) E - V_0 = aE - V_0$$
$$\mathbf{p}^\star \equiv \left(1 - \frac{c_{15}}{m^2} j_{\mathrm{d}}^{ii}\right) \mathbf{p} = b\mathbf{p}$$
$$M^\star \equiv m + S + \frac{c_{14}}{m^2} j_{\mathrm{d}}^0 E = M + \frac{c_{14}}{m^2} j_{\mathrm{d}}^0 E$$

The non-trivial solution requires the dispersion relation $\mathbf{p}^{\star 2} + M^{\star 2} = E^{\star 2}$. Positive energy:

$$E = \frac{aV_0 + j_{\mathrm{d}}^0 M + \sqrt{\left(aM + j_{\mathrm{d}}^0 V_0\right)^2 + \left[a^2 - (j_{\mathrm{d}}^0)^2\right](b\mathbf{p})^2}}{a^2 - (j_{\mathrm{d}}^0)^2}$$

Reformulate the Dirac equation:

$$\left(\Gamma^\mu i\partial_\mu - M - \gamma^0 V_0\right)\psi = 0$$

where

$$\Gamma^\mu = \gamma^\mu - \frac{c_{14}}{m^2} j_{\mathrm{d}}^\mu - \frac{c_{15}}{m^2} j_{\mathrm{d}}^{\nu\mu}\gamma_\nu$$

The truly conserved current is $J^\mu \equiv \langle \bar{\psi}\Gamma^\mu\psi \rangle$.

Normalization condition:

$$\bar{u}\Gamma^0 u = \bar{u}\left(a\gamma^0 - j_{\mathrm{d}}^0\right) u = 1$$

Positive-energy solution:

$$u = N \begin{pmatrix} 1 \\ \dfrac{\boldsymbol{\sigma}\cdot\mathbf{p}^\star}{M^\star + E^\star} \end{pmatrix} \chi$$

$$u = \sqrt{\frac{E^\star + M^\star}{2\left(aE^\star - j_{\mathrm{d}}^0 M^\star\right)}} \begin{pmatrix} 1 \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}^\star}{M^\star + E^\star} \end{pmatrix} \chi$$

A closed loop of self-consistent iteration has been formed:

$$E = \frac{aV_0 + j_{\mathrm{d}}^0 M + \sqrt{\left(aM + j_{\mathrm{d}}^0 V_0\right)^2 + \left[a^2 - (j_{\mathrm{d}}^0)^2\right](b\mathbf{p})^2}}{a^2 - (j_{\mathrm{d}}^0)^2}$$

$$u = \sqrt{\frac{E^\star + M^\star}{2\left(aE^\star - j_{\mathrm{d}}^0 M^\star\right)}} \begin{pmatrix} 1 \\ \frac{\sigma\cdot\mathbf{p}^\star}{M^\star + E^\star} \end{pmatrix} \chi$$

$$\Rightarrow$$
Iteration
$$\Leftarrow$$

$$\rho_S = 2\sum_\tau \int \frac{d^3p}{(2\pi)^3} \bar{u}(\mathbf{p})\, u(\mathbf{p})$$

$$\rho_V = 2\sum_\tau \int \frac{d^3p}{(2\pi)^3} \bar{u}(\mathbf{p})\, \gamma^0 u(\mathbf{p})$$

$$j_{\mathrm{d}}^0 = 2\sum_\tau \int \frac{d^3p}{(2\pi)^3} E(\mathbf{p})\, \bar{u}(\mathbf{p})\, u(\mathbf{p})$$

$$j_{\mathrm{d}}^{00} = 2\sum_\tau \int \frac{d^3p}{(2\pi)^3} E(\mathbf{p})\, \bar{u}(\mathbf{p})\, \gamma^0 u(\mathbf{p})$$

$$j_{\mathrm{d}}^{ii} = \frac{2}{3}\sum_\tau \sum_j \int \frac{d^3p}{(2\pi)^3} p_j \bar{u}(\mathbf{p})\, \gamma^j u(\mathbf{p})$$

Spherical Even-even Nuclei

The Lagrangian in Spherical Nuclei takes the form:

$$\mathcal{L} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{LO}} + \mathcal{L}_{\text{NLO}} + \mathcal{L}_{\text{em}}$$

Mean-field & "No-sea" approximation

$\Rightarrow$

The energy density functional

$$E_{\text{DF}} = \int \mathrm{d}^3 r \mathcal{E}(\boldsymbol{r})$$

with the energy density

$$\mathcal{E}(\boldsymbol{r}) = \mathcal{E}_{\text{kin}}(\boldsymbol{r}) + \mathcal{E}_{\text{LO}}(\boldsymbol{r}) + \mathcal{E}_{\text{NLO}}(\boldsymbol{r}) + \mathcal{E}_{\text{em}}(\boldsymbol{r})$$

where

$$\mathcal{E}_{\text{kin}}(\boldsymbol{r}) = \sum_k v_k^2 \psi_k^\dagger(\boldsymbol{r})(\boldsymbol{\alpha}\cdot\boldsymbol{p} + \beta m)\psi_k(\boldsymbol{r})$$

$$\mathcal{E}_{\text{em}}(\boldsymbol{r}) = \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - F^{0\mu}\partial_0 A_\mu + eA_\mu j_p^\mu$$

$\mathcal{O}(q^0)$	$\tilde{O}_1, \tilde{O}_{1t}$	$(\bar{\psi}\psi)(\bar{\psi}\psi), (\bar{\psi}\vec{\tau}\psi)(\bar{\psi}\vec{\tau}\psi)$	$\mathcal{O}(q^2)$	$\tilde{O}_{10}, \tilde{O}_{10t}$	$\frac{1}{4m_n^2}(\bar{\psi}\psi)\partial^2(\bar{\psi}\psi), \frac{1}{4m_n^2}(\bar{\psi}\vec{\tau}\psi)\partial^2(\bar{\psi}\vec{\tau}\psi)$
	$\tilde{O}_2, \tilde{O}_{2t}$	$(\bar{\psi}\gamma^\mu\psi)(\bar{\psi}\gamma_\mu\psi), (\bar{\psi}\vec{\tau}\gamma^\mu\psi)(\bar{\psi}\vec{\tau}\gamma_\mu\psi)$		$\tilde{O}_{11}, \tilde{O}_{11t}$	$\frac{1}{4m_n^2}(\bar{\psi}\gamma^\mu\psi)\partial^2(\bar{\psi}\gamma_\mu\psi), \frac{1}{4m_n^2}(\bar{\psi}\vec{\tau}\gamma^\mu\psi)\partial^2(\bar{\psi}\vec{\tau}\gamma_\mu\psi)$
	$\tilde{O}_3$	$(\bar{\psi}\gamma_5\gamma^\mu\psi)(\bar{\psi}\gamma_5\gamma^\mu\psi)$		$\tilde{O}_{12}$	$\frac{1}{4m_n^2}(\bar{\psi}\gamma_5\gamma^\mu\psi)\partial^2(\bar{\psi}\gamma_5\gamma_\mu\psi)$
	$\tilde{O}_4$	$(\bar{\psi}\sigma^{\mu\nu}\psi)(\bar{\psi}\sigma_{\mu\nu}\psi)$		$\tilde{O}_{13}$	$\frac{1}{4m_n^2}(\bar{\psi}\sigma^{\mu\nu}\psi)\partial^2(\bar{\psi}\sigma_{\mu\nu}\psi)$
$\mathcal{O}(q^2)$	$\tilde{O}_5$	$(\bar{\psi}\gamma_5\psi)(\bar{\psi}\gamma_5\psi)$		$\tilde{O}_{14}$	$\frac{1}{4m_n^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}i\overleftrightarrow{\partial}^\alpha\psi\right) - \tilde{O}_1$
	$\tilde{O}_6$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_5\gamma_\alpha i\overleftrightarrow{\partial}_\mu\psi\right)$		$\tilde{O}_{15}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_\mu i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_2$
	$\tilde{O}_7$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\sigma_{\mu\alpha}i\overleftrightarrow{\partial}_\nu\psi\right)$		$\tilde{O}_{16}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_5\gamma_\mu i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_3$
	$\tilde{O}_8$	$\frac{1}{4m_n^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\mu\psi\right)\partial^\nu(\bar{\psi}\sigma_{\mu\nu}\psi)$		$\tilde{O}_{17}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\sigma_{\mu\alpha}i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_4$
	$\tilde{O}_9$	$\frac{1}{4m_n^2}(\bar{\psi}\sigma^{\mu\alpha}\psi)\partial_\alpha\partial^\nu(\bar{\psi}\sigma_{\mu\nu}\psi)$			

What about $\mathcal{E}_{\text{LO}}(\boldsymbol{r})$ and $\mathcal{E}_{\text{NLO}}(\boldsymbol{r})$?

Spherical Even-even Nuclei

Define several densities and currents:

$$\rho_S = \sum_k v_k^2 \bar{\psi}_k(\boldsymbol{r})\psi_k(\boldsymbol{r})$$

$$\vec{\rho}_{tS} = \sum_k v_k^2 \bar{\psi}_k(\boldsymbol{r})\vec{\tau}\psi_k(\boldsymbol{r})$$

$$\rho_V = \sum_k v_k^2 \bar{\psi}_k(\boldsymbol{r})\gamma^0\psi_k(\boldsymbol{r})$$

$$\vec{\rho}_{tV} = \sum_k v_k^2 \bar{\psi}_k(\boldsymbol{r})\vec{\tau}\gamma^0\psi_k(\boldsymbol{r})$$

$$\rho_T = \hat{\boldsymbol{r}}\cdot\sum_k v_k^2 \bar{\psi}_k(\boldsymbol{r})\gamma^0\boldsymbol{\gamma}\psi_k(\boldsymbol{r})$$

$$\rho_{SE} = \sum_k v_k^2\epsilon_k \bar{\psi}_k(\boldsymbol{r})\psi_k(\boldsymbol{r})$$

$$\rho_{VE} = \sum_k v_k^2\epsilon_k \bar{\psi}_k(\boldsymbol{r})\gamma^0\psi_k(\boldsymbol{r})$$

$$\rho_{TE} = \hat{\boldsymbol{r}}\cdot\sum_k v_k^2\epsilon_k \bar{\psi}_k(\boldsymbol{r})\gamma^0\boldsymbol{\gamma}\psi_k(\boldsymbol{r})$$

$$J_{\mathrm{d}}^{ij} = T_R(\boldsymbol{r})\hat{r}^i\hat{r}^j + T_T(\boldsymbol{r})\left(\delta^{ij}-\hat{r}^i\hat{r}^j\right)$$

$$J_{\mathcal{T}} = -\frac{1}{r}\sum_k v_k^2\bar{\psi}_k\boldsymbol{\Sigma}\cdot\boldsymbol{L}\psi_k$$

Relativistic NN contact Lagrangian up to NLO and their contributions to energy density functional:

Order	$\tilde{O}_i$		$\mathcal{E}_i$
$\mathcal{O}(q^0)$	$\tilde{O}_1, \tilde{O}_{1t}$	$(\bar{\psi}\psi)(\bar{\psi}\psi), (\bar{\psi}\vec{\tau}\psi)(\bar{\psi}\vec{\tau}\psi)$	$\rho_S^2, \vec{\rho}_{tS}^2$
	$\tilde{O}_2, \tilde{O}_{2t}$	$(\bar{\psi}\gamma^\mu\psi)(\bar{\psi}\gamma_\mu\psi), (\bar{\psi}\vec{\tau}\gamma^\mu\psi)(\bar{\psi}\vec{\tau}\gamma_\mu\psi)$	$\rho_V^2, \vec{\rho}_{tV}^2$
	$\tilde{O}_3$	$(\bar{\psi}\gamma_5\gamma^\mu\psi)(\bar{\psi}\gamma_5\gamma^\mu\psi)$	0
	$\tilde{O}_4$	$(\bar{\psi}\sigma^{\mu\nu}\psi)(\bar{\psi}\sigma_{\mu\nu}\psi)$	$-2\rho_T^2$
$\mathcal{O}(q^2)$	$\tilde{O}_5$	$(\bar{\psi}\gamma_5\psi)(\bar{\psi}\gamma_5\psi)$	0
	$\tilde{O}_6$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_5\gamma_\alpha i\overleftrightarrow{\partial}_\mu\psi\right)$	0
	$\tilde{O}_7$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\sigma_{\mu\alpha}i\overleftrightarrow{\partial}_\nu\psi\right)$	$-\frac{1}{m^2}\rho_{TE}^2 - \frac{1}{2m^2}J_{\mathcal{T}}^2$
	$\tilde{O}_8$	$\frac{1}{4m_n^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\mu\psi\right)\partial^\nu\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$	$\frac{1}{2m^2}\rho_{SE}\nabla\cdot(\hat{\boldsymbol{r}}\rho_T)$
	$\tilde{O}_9$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\alpha}\psi\right)\partial_\alpha\partial^\nu\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$	$-\frac{1}{4m^2}\left[\nabla\cdot(\hat{\boldsymbol{r}}\rho_T)\right]^2$
	$\tilde{O}_{10}, \tilde{O}_{10t}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\psi\right)\partial^2\left(\bar{\psi}\psi\right), \frac{1}{4m_n^2}\left(\bar{\psi}\vec{\tau}\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\psi\right)$	$-\frac{1}{4m^2}\rho_S\Delta\rho_S, -\frac{1}{4m^2}\vec{\rho}_{tS}\Delta\vec{\rho}_{tS}$
	$\tilde{O}_{11}, \tilde{O}_{11t}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\gamma_\mu\psi\right), \frac{1}{4m_n^2}\left(\bar{\psi}\vec{\tau}\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\vec{\tau}\gamma_\mu\psi\right)$	$-\frac{1}{4m^2}\rho_V\Delta\rho_V, -\frac{1}{4m^2}\vec{\rho}_{tV}\Delta\vec{\rho}_{tV}$
	$\tilde{O}_{12}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu\psi\right)\partial^2\left(\bar{\psi}\gamma_5\gamma_\mu\psi\right)$	0
	$\tilde{O}_{13}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}\psi\right)\partial^2\left(\bar{\psi}\sigma_{\mu\nu}\psi\right)$	$-\frac{1}{4m^2}\rho_T\Delta\rho_T$
	$\tilde{O}_{14}$	$\frac{1}{4m_n^2}\left(\bar{\psi}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_1$	$\frac{1}{m^2}\rho_{SE}^2 - \rho_S^2$
	$\tilde{O}_{15}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_\mu i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_2$	$\frac{1}{m^2}\rho_{VE}^2 + \frac{1}{4m^2}\left(T_R^2 + 2T_T^2\right) - \rho_V^2$
	$\tilde{O}_{16}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\gamma_5\gamma^\mu i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\gamma_5\gamma_\mu i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_3$	0
	$\tilde{O}_{17}$	$\frac{1}{4m_n^2}\left(\bar{\psi}\sigma^{\mu\nu}i\overleftrightarrow{\partial}^\alpha\psi\right)\left(\bar{\psi}\sigma_{\mu\nu}i\overleftrightarrow{\partial}_\alpha\psi\right) - \tilde{O}_4$	$-\frac{2}{m^2}\rho_{TE}^2 - \frac{1}{m^2}J_{\mathcal{T}}^2 + \rho_T^2$

$\mathcal{E}_{\mathrm{LO}}(\boldsymbol{r})$ and $\mathcal{E}_{\mathrm{NLO}}(\boldsymbol{r})$ are supposed to be linear combination of $\mathcal{E}_i$.

Spherical Even-even Nuclei

Total energy for the Spherical even-even nuclei:

$$E_{\mathrm{tot}} = E_{\mathrm{DF}} + E_{\mathrm{pair}} + E_{\mathrm{c.m.}}^{\mathrm{mic}}$$

- The EDF for Spherical even-even nuclei system

$$E_{\mathrm{DF}} = \int \mathrm{d}^3 r \left[\mathcal{E}_{\mathrm{kin}}(\boldsymbol{r}) + \mathcal{E}_{\mathrm{LO}}(\boldsymbol{r}) + \mathcal{E}_{\mathrm{NLO}}(\boldsymbol{r}) + \mathcal{E}_{\mathrm{em}}(\boldsymbol{r})\right]$$

- The pairing energy

$$E_{\mathrm{pair}} = -\frac{1}{2}\sum_{\tau=n,p} \mathrm{Tr}\left(\Delta_\tau \kappa_\tau^*\right),$$

  The pairing correlations are described by the Tian-Ma-Ring finite-range separable pairing force

$$V^{pp} = -G\,\delta(\boldsymbol{R} - \boldsymbol{R}')P(\mathbf{r})P(\mathbf{r}')\,\frac{1}{2}(1 - P^\sigma),$$

  where $P(r) = \frac{1}{(4\pi a^2)^{3/2}} \exp\left(-\frac{r^2}{4a^2}\right)$, $G = 728\ \mathrm{MeV\,fm^3}$, $a = 0.644\ \mathrm{fm}$.

- The microscopic center of mass correction energy

$$E_{\mathrm{c.m.}}^{\mathrm{mic}} = -\frac{1}{2mA}\left\langle \hat{\boldsymbol{P}}_{\mathrm{c.m.}}^2 \right\rangle$$

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Numerical Details

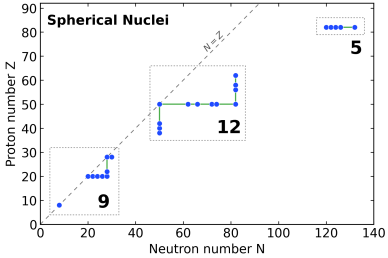
- **Two objects:**
  - Nuclear matter quantities: $\rho_0, E/A, E_{\rm sym}, K_\infty, L$.
  - E_B and $r_{\rm ch}$ of selected 26 spherical even-even nuclei.
- **The squared deviation:**

$$\chi^2(\boldsymbol{c}) = \frac{w_1}{N_A}\sum_{i=1}^{N_A}\left[\frac{O_i^{\rm exp}-O_i^{\rm cal}(\boldsymbol{c})}{\sigma_E}\right] + \frac{w_1}{N_A}\sum_{i=1}^{N_A}\left[\frac{O_{r_{ch}i}^{\rm exp}-O_{r_{ch}i}^{\rm cal}(\boldsymbol{c})}{\sigma_{r_{ch}}}\right] + \frac{w_2}{N_{\rm NM}}\sum_{i=1}^{N_{\rm NM}}\left[\frac{O_{{\rm NM}i}^{\rm exp}-O_{{\rm NM}i}^{\rm cal}(\boldsymbol{c})}{\sigma_{{\rm NM}i}}\right]$$

- **Fitting strategy:**
  - Bulk first: Nuclear matter constrains c_1, c_2, c_{1t}, c_{2t}.
  - Tensor response: Spherical nuclei constrain c_4.
    $\Rightarrow$ LO parameters: $c_1, c_2, c_4, c_{1t}, c_{2t}$
  - Scalar and vector derivative terms: Release c_{14}, c_{15}.
  - Resolve the surface: Spherical nuclei constrain $c_{10}, c_{11}, c_{13}, c_{10t}, c_{11t}$.
  - Final global optimization: Refit all identifiable parameters.
    $\Rightarrow$ Up to NLO parameters:
    $c_1, c_2, c_4, c_{1t}, c_{2t}, c_7, c_8, c_9, c_{10}, c_{11}, c_{13}, c_{14}, c_{15}, c_{17}, c_{10t}, c_{11t}$

Quantity	Empirical	σ
ρ_0 (fm^{-3})	0.160	0.01
E/A (MeV)	-15.86	0.57
K_∞ (MeV)	240	20(500)
$E_{\rm sym}$ (MeV)	31.60	2.66
L (MeV)	106	37

Table: Nuclear matter properties[3].



[3] X. Zhang, C. C. Wang, C. R. Ding, and J. M. Yao, Phys. Rev. C 112, L021302 (2025)

Numerical Details

- **Parameters and their naturalness:**
  The QCD-based Lagrangian

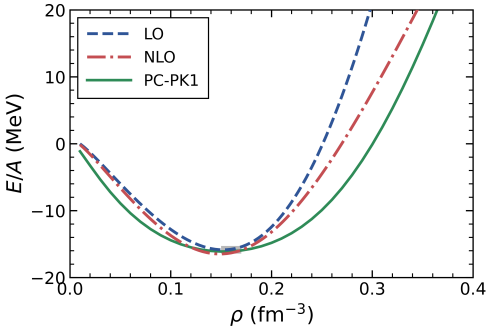
$$\mathcal{L} \sim -c_{ln}\left[\frac{\bar{\psi}\psi}{f_\pi^2\Lambda}\right]^l\left[\frac{\partial^\mu}{\Lambda}\right]^n f_\pi^2\Lambda^2$$

where $f_\pi = 92.5\mathrm{MeV}, \Lambda = 770\mathrm{MeV}$.

Parameter [fm^2]	LO		NLO	
	Value	c_{ln}	Value	c_{ln}
c_1	−15.399844	−1.691986	−15.792378	−1.735114
c_2	11.751073	1.291094	12.002372	1.318705
c_4	−9.140671	−1.004289	−20.125186	−2.211161
c_{1t}	−0.096960	−0.042612	−0.075427	−0.033149
c_{2t}	0.836870	0.367790	1.010016	0.443884
c_7			−6.618368	−0.122242
c_8			83.949069	1.550548
c_9			279.213617	5.157103
c_{10}			348.586025	6.438419
c_{11}			−244.183998	−4.510103
c_{13}		–	−19.502227	−0.360208
c_{14}			7.124069	0.131582
c_{15}			−6.980556	−0.128932
c_{17}			−8.341535	−0.154069
c_{10t}			3.126189	0.230965
c_{11t}			16.374705	1.209774

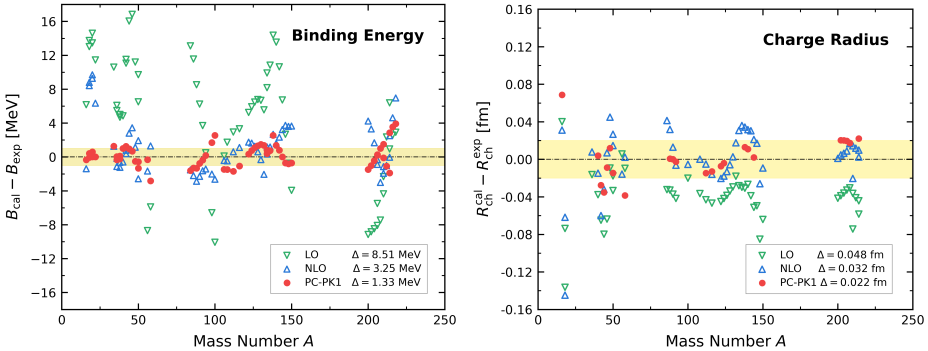
Numerical Details

- **EoS and Nuclear matter quantities**



Quantity	LO	NLO	Empirical	PC-PK1
ρ_0 (fm^{-3})	0.152	0.149	0.160(0.10)	0.154
E/A (MeV)	-15.86	-16.48	-15.86(0.57)	16.12
M^*/M	0.54	0.35	–	0.59
K_∞ (MeV)	550	498	240(20)	238
$E_{\rm sym}$ (MeV)	31.54	30.86	31.60(266)	35.6
L (MeV)	105.8	91	106(37)	113

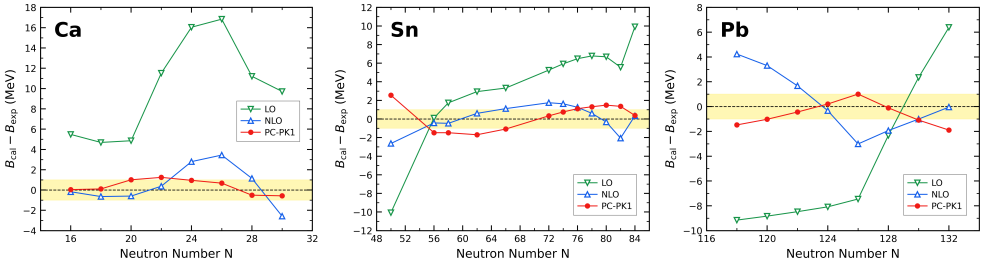
- **Deviations of binding energies and charge radius between theoretical and experimental values**



NLO reduces the rms deviations from 8.51 MeV to 3.25 MeV for binding energies and from 0.048 fm to 0.032 fm for charge radii, showing a significant global improvement over LO.

Outline

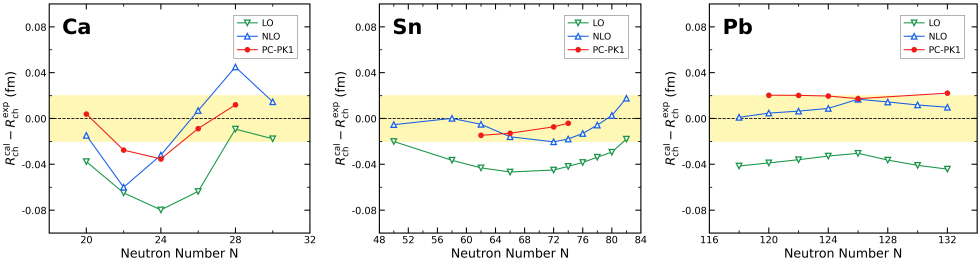
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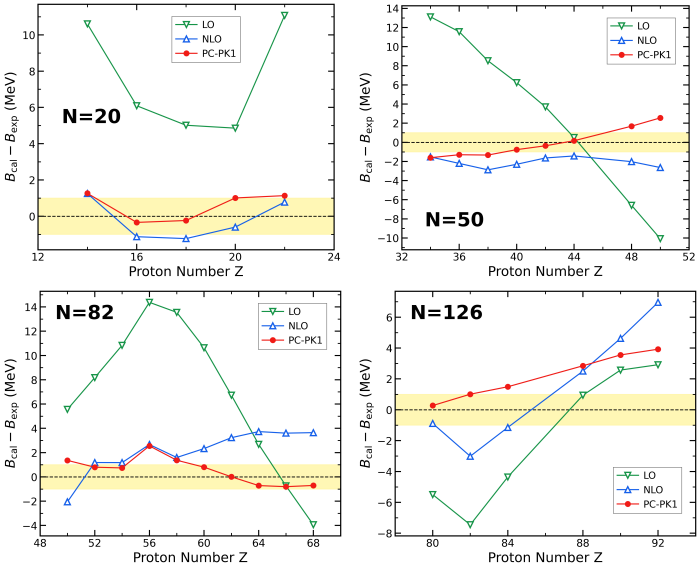
NLO largely removes the strong systematic trends of the LO residuals along the Ca, Sn, and Pb isotopic chains, indicating an improved description of the isospin dependence.

The improvement is particularly pronounced for Sn, while local discrepancies remain around shell-sensitive regions.

Results and Discussion



Results and Discussion



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Summary

- Based on the relativistic chiral NN contact interaction, a covariant energy density functional has been constructed up to NLO.
  - The fitted couplings are generally consistent with QCD naturalness.
  - NLO significantly improves the description of binding energies and charge radii of finite nuclei
  - Systematic trends along isotopic and isotonic chains are largely reduced, indicating improved surface and isovector properties.

Thanks for your attention!